CS 3510: Design & Analysis of Algorithms (Spring 2026)

Lectures: Section A: Tuesday & Thursday 12:30 – 1:45 am in Howey Physics L4.
Section B: Tuesday & Thursday 9:30 – 10:45 am in Instructional Center 103.

Instructor: Will Perkins
Email: wperkins3@gatech.edu
Office hours: Klaus 2222A, Tuesdays/Thursdays, 1:50–2:50 (informal office hours for quick questions on the walk from the lecture hall to Klaus after class)

Teaching Assistants: We have a great team of graduate and undergraduate TA's (shared with my second section of 3510). Their office hours will be posted on Canvas.

For a large class like this, the TA's have a ton of work to do (grading homeworks and exams, holding office hours, answering discussion board questions, printing and scanning exams). Many of the course policies are designed to ameliorate this workload; please also be polite and considerate of their time.

Course webpage: All course materials and announcements will be posted on Canvas. It is your responsibility to check Canvas for assignments, announcements, and review material.
We will use a Ed Discussion site here: https://edstem.org/us/courses/91955/ (also linked from Canvas) for class discussions, homework questions, questions about lectures and more. Please always ask your questions on the discussion board rather than by email. The discussion board is meant for questions about content, clarification of course logistics. Questions off-topic and those that reveal too much about homework solutions may be deleted.

Course description
This is a course in the fundamentals of algorithm design and analysis. Algorithms are ubiquitous and govern the modern world through data analysis, search, machine learning, and cryptography. The ability to design correct and efficient algorithms is the key to unlocking their power. The mathematical analysis of algorithms provides the rigorous foundation needed to evaluate and compare different algorithms' performance and scalability to large instances. By the end of this course, you will have a strong foundation in the principles of algorithm design and the ability to analyze their performance.

Topics

(1) Asymptotic analysis of algorithms; divide and conquer algorithmic paradigm; algorithms for numbers; RSA cryptosystem
(2) Graph algorithms; shortest paths, depth-first search, breadth-first search; matchings
(3) Dynamic programming; longest common subsequence; all-pairs shortest paths; Markov decision processes
(4) Computational complexity and NP-completeness; P vs NP; NP-hardness reductions; TSP; Max-Cut; approximation algorithms.

After these four main topics we will have an additional three lectures (time permitting) on some advanced topics including sublinear and randomized algorithms.

Textbook *Algorithms* by S. Dasgupta, C. Papadimitriou, and U. Vazirani (required).

Other useful resources

- Handwritten lecture notes will be posted after class on the Ed Discussion site.
- The textbook *Algorithm Design* by Kleinberg and Tardos
- Short videos from Jake Abernethy are available at https://gt-cs-3510.github.io/

Grading

Grading for the course is divided into two components:

- Exams 80%; we will have 4 in-class exams plus the final exam. The lowest exam score of the 5 will be dropped.
- Homework 15%; we will have 9 homework assignments; the lowest two scores will be dropped. Only one or two selected questions will be graded. Homework grading will be very lenient (you will receive half credit simply for completing the assignment). There will be no regrade requests for homeworks. There will be no extensions for homeworks (if you are sick, you can use one of the two dropped grades).
- Attendance 5%; attendance in lectures is mandatory. While we will not record attendance regularly, if class attendance drops below 75% for multiple lectures, we will start taking attendance and reflecting it in the attendance portion of the grade. If attendance is always above 75% all students will receive full attendance points.

Your average will determine your final grade:

- $90 - 100$: A
- $80 - 89$: B
- $70 - 79$: C
- $60 - 69$: D
- $0 - 59$: F

These ranges may be adjusted later (i.e., a grade curve) but only in your favor. There will be no CIOS bonus (it doesn't make much sense if there is a grade curve).

Assignments

- **Homework** will be assigned 2 homeworks per unit with one more after the 4th exam. You will have roughly a week to complete the homework. You may discuss homework with other students in the class, but when you write it up you must do it all alone. You must write the names of students you worked with on your homework. You may look up reference material online and in textbooks to help with your homework but you may not look up or copy answers to homework questions. We will drop your two lowest homework grades.

  Homework must be typewritten, handwritten solutions will not be accepted. Using Latex (via Overleaf or otherwise) is strongly recommended.
- **In-class exams** will occur during the normally scheduled class time on the following dates:
  - January 29
  - February 19
  - March 17
  - April 14

  You will receive review sheets before each test detailing what will be covered on that test.
- There is cumulative **final exam**. Your exam score will be the average of the 4 best scores from your 4 in-class exams and 1 final exam.

Succeeding in the course

There are many resources available that cover the course material: the lectures, the textbook, lecture notes from this and other courses, lecture videos from other courses. It often is helpful to hear more than one perspective on a topic and so I encourage you to seek out other resources and see different explanations of the concepts we learn in class.

While you can catch up on material from a missed lecture by reading the textbook, I strongly recommend attending all lectures: the lectures will give you an understanding of how things

fit together, what is hard and what is easy, and the intuition behind the concepts. From past experience it is very clear that students who attend lecture regularly perform best in the class.

Algorithm design and analysis is a topic at the intersection of computer science and mathematics, and it has its own language: the language of mathematics and pseudocode. As with any language, the only way to master it is to practice! This is why doing the homeworks, doing sample problems, do practice tests is critically important - this is by far the best way to learn.

Course Policies

- Late homework: late homework not accepted (for sickness and emergencies, use the two dropped HW grades).
- Missed exams: if you miss an exam due to illness or other circumstances with a valid doctor's note or other official notice, you will take a make-up exam; if you are absent without a note, you will get a 0 and will need to use your one dropped exam score.
- Regrades: there are no regrades on homeworks. You can submit a maximum total of 3 regrade requests on midterm exams; these are meant to be used for clear mistakes in grading, not for judgment of partial credit. You will only receive credit for the regrade if your answer is completely correct but was graded in error.
- Attendance in lecture is mandatory; see above.
- Class discussion, working with each other, and communicating with fellow students, the instructor, and the TAs are all essential elements of the course. I expect all of us to treat each other with respect and courtesy in all of our interactions.
- Please read Georgia Tech's guide to Student–Faculty Expectations here: https://catalog.gatech.edu/rules/21/
- If any issues arise that will prevent you from attending class, completing an assignment, or enjoying and thriving in the class, please let me know as soon as possible so we can work together to find a solution. It is always helpful if you can present a possible solution to a problem when you bring it to me.
- Academic honesty
  - Exams will be taken during class time in the lecture hall. Since the room is crowded it is very important that you do not look at other students' work during the exams. I am teaching two sections of 3510 on the same days and following the same schedule. While the exams will have different questions, it is forbidden to discuss the questions of the exam with students from the other section before both exams are finished.
  - You may work together on homework, but each student must write their own solutions (not copy another student's answer). You are forbidden from searching for homework solutions online or using LLM's to answer homework questions for you (see below for other uses of LLM's).
  - Please read the Georgia Tech Honor Code here: https://policylibrary.gatech.edu/student-life/academic-honor-code. Penalties for academic misconduct include getting a zero for an assignment or for the course, see here: https://osi.gatech.edu/process/academic-misconduct-sanctioning-guidelines
- Use of AI / LLM's / chatGPT: LLM's can quite likely answer most or all of the homework questions. You are not allowed to use them to solve your homework. It will be difficult for us to check this, but it is not in your interest to do this; for one, homeworks will be graded leniently; more importantly, the exercise of thinking through homework problems is one of the key ways you will learn and develop your algorithmic thinking in this course. This will help you not only get good scores on the exams but also accomplish the main goal of the course: leave you with deep and lasting knowledge about algorithm design. You should not upload the questions or your solutions to an AI/LLM site for any reason (e.g. for putting written solutions into Latex code).

On the other hand, I do encourage you to experiment with using LLM's as a personalized tutor to help you understand concepts in the course. For instance, asking chatGPT to give you an interactive quiz based on the topic list for an exam is a great idea. I will discuss this more before the first exam.

Disability policy

Students with disabilities who require accommodations for access and participation in this course should register with the Office of Disability Services (ODS). Please contact ODS here: https://disabilityservices.gatech.edu/

Tentative course schedule (subject to change; review classes might be dropped if we are behind schedule on the main material):

Jan 13 Big-Oh notation, Bubble Sort, Fibonacci numbers (HW 1 released)
Jan 15 Divide and Conquer paradigm: Integer Multiplication, Merge Sort, and Recurrences
Jan 20 Matrix Multiplication, Fast Fourier Transform (HW 2 released)
Jan 22 Fast Fourier Transform, RSA cryptography
Jan 27 Finish RSA, Review
Jan 29 Exam 1

Feb 3 Graphs; connected components; DFS (HW 3 released)
Feb 5 DFS in digraphs; DAG's
Feb 10 Shortest paths; BFS (HW 4 released)
Feb 12 Kruskal's algorithm, Prim's algorithm
Feb 17 Greedy algorithms: Huffman coding, Set Cover
Feb 19 Exam 2

Feb 24 Dynamic Programming (HW 5 released)
Feb 26 Independent set in trees, Knapsack
Mar 3 All-pairs shortest paths (HW 6 released)
Mar 5 Traveling Salesman Problem, Chain Matrix Multiplication
Mar 10 Markov decision processes, Review
Mar 12 Introduction to Complexity
Mar 17 Exam 3

Mar 19 P vs NP (HW 7 released)
Mar 31 NP-hardness reductions
Apr 2 Approximation algorithms I (HW 8 released)
Apr 7 Approximation algorithms II
Apr 9 Review
Apr 14 Exam 4

Apr 16 Sublinear algorithms
Apr 21 Randomized algorithms I (HW 9 released)
Apr 23 Randomized algorithms II
Apr 28 Review

Final Exam : as scheduled on final exam martix: https://registrar.gatech.edu/public/files/202602%20Final%20Exam%20Matrix_0.pdf

- 9:30 am section exam is on Monday May 4, at 8:00am
- 12:30 pm section exam is on Thursday May 7, 11:20 am